



Name.....

Roll No.....

ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.*(An ISO 9001:2015 Certified College)***DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING****B. TECH DEGREE FIRST INTERNAL TEST- OCTOBER 2023****Seventh Semester (AD, CE, CS, ME)****ECT 425: BIOMEDICAL INSTRUMENTATION (Open Elective)****Time: 2 Hours****Max. Marks: 50****PART A**

Answer all questions (3 x 5 = 15 Marks)

Qn.no.	Questions	Marks	K level	CO & KL
✓1	Draw the structure of the heart showing the major components.	3	K2	CO1 & K2
✓2	Define ECG ,EEG ,EMG.	3	K1	CO1 & K2
✓3	Draw and briefly explain Einthoven Triangle	3	K2	CO2 & K2
✓4	Explain half-cell potential?	3	K1	CO2 & K2
✓5	Define systolic and diastolic pressure?	3	K2	CO3 & K2

PART B

Answer five questions (35 Marks)

Module 1: Qn. No 6,7 and 8 – Answer any 2

Module 2: Qn. No 9, 10 ,11– Answer any 2

Module 3: Qn. No 12 is Mandatory

Qn no.	Question	Marks	K level	CO&KL
✓6	Describe in detail the formation of resting and action potential in human body	7	K2	CO1 & K2
7	Draw the general block diagram of a biomedical instrumentation system and explain the functions of each block	7	K2	CO1 & K2
✓8	With neat sketch describe the functioning of human respiratory system	7	K2	CO1 & K2
9	Draw the block diagram of ECG machine and explain the functions of each block	7	K2	CO2 & K2
✓10	Describe electro conduction system of heart	7	K2	CO2 & K2
11	Write short notes on bio potential electrode? With neat sketches explain needle & micro electrodes.	7	K2	CO2 & K2
✓12	With the help of neat diagram explain oscillometric method helps to measure blood pressure	7	K2	CO3 & K2



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ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.*(An ISO 9001:2015 Certified College)***DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING****B. TECH DEGREE 12th INTERNAL TEST- DECEMBER 2023****Seventh Semester (AD, CE, CS, ME)****ECT 425: BIOMEDICAL INSTRUMENTATION (Open Elective)****Time: 2 Hours****Max. Marks: 50****PART A**

Answer all questions (3 x 5 = 15 Marks)

Qn no.	Questions	Marks	K level	CO & KL
1	Write short notes on phonocardiography	3	K2	CO3 & K2
2	What is lithotripsy	3	K1	CO4 & K2
3	Write short notes on standard 10-20 electrode placement system for EEG	3	K2	CO4 & K2
4	Explain diathermy?	3	K1	CO5 & K2
5	Explain generation of Xrays	3	K2	CO5 & K2

PART B

Answer five questions (35 Marks)

Module 4: Qn. No 6,7 and 8 – Answer any 2**Module 5: Qn. No 9, 10, 11– Answer any 2****Module 3: Qn. No 12 is Mandatory**

Qn no.	Question	Marks	K level	CO&KL
6	With the help of neat sketch explain the working of heart lung machine	7	K2	CO4 & K2
7	Explain the process involved in hemodialysis with the help of diagram	7	K2	CO4 & K2
8	What is pacemakers? Explain the types of pacemakers?	7	K2	CO4 & K2
9	What is the basic principle of CT? How image construction is done in CT?	7	K2	CO5 & K2
10	Draw & explain the principle of ultrasound imaging?	7	K2	CO5 & K2
11	Explain different methods of electric accident prevention?	7	K2	CO5 & K2
12	What is plethysmography? Explain photoelectric plethysmography with neat diagram?	7	K2	CO3 & K2

Reg No.: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Seventh Semester B.Tech Degree Regular and Supplementary Examination December 2023 (2019 Scheme)

Course Code: ECT425**Course Name: BIOMEDICAL INSTRUMENTATION****Max. Marks: 100****Duration: 3 Hours****PART A***Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|--|-----|
| ✓1 | Differentiate between resting potential and action potential. | (3) |
| ✓2 | Enumerate different EEG rhythms with frequency ranges. | (3) |
| ✓3 | Draw and explain an ECG waveform. | (3) |
| ✓4 | Write short notes on surface electrodes. | (3) |
| ✓5 | Differentiate between systolic and diastolic pressure. | (3) |
| ✓6 | What is phonocardiography? | (3) |
| 7 | Draw a neat plot showing different respiratory parameters. | (3) |
| 8 | What is a cardiac defibrillator? | (3) |
| ✓9 | What is the difference between a conventional X-ray and computer tomography? | (3) |
| 10 | Mention any three applications of telemetry medicine. | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- 11 a) What is the need for a biomedical instrument? With a neat block diagram explain the significance of each basic component in it. (7)
- b) Discuss how bioelectric potentials are developed inside human body. (7)

OR

- 12 a) With suitable sketch explain the functioning of human respiratory system. (6)
- b) Explain in detail about the problems encountered in a biomedical measuring system. (8)

Module II

- 13 a) Explain any two passive transducers used in biopotential measurement. (8)

- b) Explain chopper amplifier with a neat circuit diagram. (6)

OR

- 14 a) With neat diagrams explain the 12 lead systems in ECG measurement. What is the significance of Einthoven triangle? (14)

Module III

- 15 a) With neat diagram explain the ultrasonic method for blood pressure measurement. (8)

- b) Give a brief description on impedance plethysmography. (6)

OR

- 16 a) Explain how phonocardiography is used for heart sound measurement. (6)

- b) With neat diagram explain the working of an electromagnetic blood flow meter. (8)

Module IV

- 17 a) With neat block diagram explain the working of an EEG recording system. (9)

- b) Explain the working of biomedical instrumentation system used for the measurement of nerve conduction velocity. (5)

OR

- 18 a) Explain heart lung machine with the help of neat diagram. (7)

- b) With neat sketch explain how hemodialysis is performed. (7)

Module V

- 19 a) Explain the principle of CT using necessary diagrams. (6)

- b) How X-rays are produced? What are its properties? (8)

OR

- 20 a) Draw the block diagram of an ultrasound imaging system and explain its principle of operation. (8)

- b) What are the biological effects of NMR imaging over CT? (6)
